

Evolution of protease inhibitor resistance in the *gag* and *pol* genes of HIV subtype G isolates

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Received 4 September 2009; returned 9 February 2010; revised 15 March 2010; accepted 28 March 2010

Objectives: To analyse HIV Gag cleavage site (CS) and non-CS mutations in HIV non-B isolates from patients failing antiretroviral therapy.

Patients and methods: Twenty-one HIV isolates were obtained from patients infected with HIV subtype G during an outbreak in Russia 20 years ago. Most patients were failing antiretroviral therapy when genotyping was performed.

Results: HIV Gag CS mutations accumulated in protease inhibitor (PI)-resistant HIV isolates and were correlated with the presence of three or more PI resistance mutations. Only 1 of 11 HIV isolates carrying major protease mutations did not harbour treatment-associated CS mutations. Natural polymorphism 453T, often found in HIV non-B subtypes, seems to favour the selection of CS mutation 453I rather than treatment-associated CS mutation 453L. Resistance-associated non-CS mutations (123E and 200I) were also observed in PI-resistant clinical isolates. Non-CS mutations in the frameshift-regulating site, which controls the synthesis of Gag–Pol, did not affect frameshift efficiency in dual luciferase assays. Of note, one of four HIV isolates from patients failing PI therapies without protease mutations harboured Gag mutations associated with PI resistance (123E and 436R) and reverse transcriptase inhibitor mutations conferring resistance to the backbone drug.

Conclusions: HIV Gag CS mutations commonly occurred in HIV isolates from patients failing PI therapies and natural polymorphisms at the same position influence their emergence. Non-CS mutations previously associated with PI resistance were also observed in clinical isolates. Gag mutations might indicate the evolution of PI resistance even in the absence of protease mutations.

Keywords: non-B subtypes, Gag cleavage site mutations, Gag non-cleavage site mutations

Introduction

Highly active antiretroviral therapy (HAART) effectively hinders disease progression in HIV-infected patients. The emergence of HIV drug resistance in protease (PR) and reverse transcriptase (RT) is one major obstacle to achieve sustained viral suppression. Recently, it was shown that PR as well as cleavage site (CS) mutations contribute to PR inhibitor (PI) resistance.¹ Moreover, non-CS Gag mutations were also found in phenotypically PI-resistant HIV isolates without PI resistance mutations,^{2–4} even though the mechanism of action remains unclear.

Initially, it was speculated that HIV Gag mutations might be responsible for the failure of PI therapies without PR mutations. So far, PI failures solely attributed to Gag mutations have not been reported.⁵ HIV CS mutations were found to confer

phenotypic resistance ranging from a 3- to 6-fold increase in resistance factors to PIs⁶ and/or to enhance PI resistance conferred by PR mutations.^{3,7} In this study we were able to analyse HIV isolates related to an outbreak in Russia, which were previously classified as subtype G and displayed high sequence homology.⁸ The same origin of these viruses can help to identify treatment-associated mutations in contrast to natural polymorphisms. Moreover, these HIV isolates may also provide further insights into the evolution of PI resistance in non-B isolates.

Patients and methods

All patients ($n=21$) were infected with HIV-1 during an outbreak in Rostov on Don (Russian Federation) between the years 1988 and 1990.

These patients acquired nosocomial HIV infection during childhood.⁸ Plasma samples from 21 patients were obtained in the years 2002–06 during therapy failure (defined as HIV detection under antiretroviral therapy) and analysed in the Pol region as well as in the complete Gag region. Antiviral therapy regimens were mainly composed of three antiretroviral drugs but non-standard dual therapies or monotherapies were also used in single patients for different reasons.

RT-PCR amplification and sequencing of the viral PR and RT from plasma were performed using the HIV ViroSeq genotyping system (Abbott) or an in-house RT-PCR. The complete gag gene was amplified using LTR-A-f (nt 687–706: CGACGCAGGACTCGGCTTGC) and C-gag1-r (nt 2382–2404: CCAATCCCTATCATTITTTGG), and sequenced (GenBank ID GU991863–GU991883). gag sequences were assembled and edited using the Seqman software module (Lasergene) and translated thereafter into amino acid sequences. The sequences were aligned and a consensus sequence from these related HIV isolates was built. Amino acid substitutions were identified according to differences in comparison with the Russian consensus sequence.

HIV frameshift efficiency was measured with a dual luciferase reporter plasmid as previously described.⁹ The frameshift-regulating site of clinical HIV isolates was amplified [p6-4-fs-KpnI (nt 2025–2059, GGGCTGTTGGTACCTGGAAGGAAGGACACCAATGA) and p6-5-fs-BamHI (nt 2139–2170, GGGCTGTTGGATCCGGCTGTTGGCTGCTGCTC)] and introduced into the dual luciferase vector. Frameshifting was monitored by transient transfection in C33a cells, which were grown for 48 h before being harvested. Frameshift efficiencies were determined by dividing the ratio of the firefly to *Renilla* luciferase activities from the clinical isolates by the firefly to *Renilla* luciferase ratio from the in-frame pDual-HIV(0) control construct.

Statistical significance was determined using Fisher's exact test or *t*-test. Significance was considered at the 95% confidence level (*P* values <0.05).

Results

HIV Gag CS mutations significantly accumulated in HIV isolates with PI resistance mutations (*P*<0.05). In particular, HIV CS mutations 128I, 430R, 436R, 431V 451S, 452G and 453I were associated with major PR mutations, whereas CS mutations 128A, 132F, 376V and 428D were also detected in HIV isolates that had not been exposed to PIs. The number of CS mutations was significantly higher in HIV isolates with ≥ 3 PI resistance PR mutations compared with HIV isolates with 1 or 2 PI resistance mutations (1–2 PR mutations 2.2 ± 0.7 versus ≥ 3 PR mutations 5.3 ± 1.1 ; *P*<0.001) (Table 1). Only one HIV isolate (patient 19) with >3 PR mutations did not harbour one of the most frequent CS mutations (128I, 431V, 436R and 453I) and did not show changes in comparison with a genotype obtained 6 months earlier. In two patients, new PR and CS mutations developed during PI failure (patient 17: PR, 46I, 54V and 82A; and Gag, 128I, 375T, 431V and 452G; and patient 14: PR, 82S; and Gag, 436R). Additionally, HIV Gag non-CS mutations were detected in HIV isolates from patients failing PI therapy (123E, *n*=1; and 369H, *n*=1) or in PI-resistant HIV isolates (123E, *n*=3; 200I, *n*=1; and 369H, *n*=3). Within the frameshift-regulating site p1, unique Gag non-CS mutations were found in one PI-resistant HIV isolate (441R and 442R) and in one isolate without PR mutations from a patient failing PI therapy (441R).

Therefore, we analysed the frameshift efficiency of specific Russian sequences harbouring different p1 mutations. The majority of Russian sequences had p1 mutations 441N and 443E in contrast to the HxB2 reference strain, which did not affect the

frameshift efficiency (Figure 1). Moreover, mutations 440F, 441S, 441H, 441R, 442R and 449I found in a few Russian HIV isolates from patients failing either PI or RT inhibitor (RTI) therapies also did not change the frameshift efficiency of these isolates.

PI failure without PR mutations was observed in four patients (patients 10, 13, 18 and 20) (Table 1). In two patients (patients 13 and 18) the mutation patterns in Gag did not differ from other Russian sequences (RT mutations: patient 13, 67N, 69N, 103N, 215L and 219Q; and patient 18, 69S and 219Q). In the third case (patient 10) unique CS mutations were detected as well as non-CS mutations (76K, 79F, 81A and 441R) and RT mutations (181CY and 184VM), which might indicate compliance or resistance problems. The HIV isolate from patient 20 harboured non-nucleoside RTI resistance mutations while the patient was receiving only efavirenz and indinavir as antiretroviral therapy. Moreover, this HIV isolate additionally carried CS (436R) and non-CS (123E) mutations (RT mutation, 103N).

Discussion

HIV Gag CS mutations accumulated in PI-resistant HIV subtype G isolates and were even more prominent in HIV isolates harbouring ≥ 3 PI resistance PR mutations. These results were in line with previous findings in HIV subtype B isolates.¹⁰ HIV Gag CS mutation 431V was the most frequent CS mutation in these subtype G isolates, which is in accordance with previous studies analysing HIV subtype B.¹⁰ The occurrence of 431V after or during treatment failure of lopinavir therapy regimens is in line with phenotypic resistance data demonstrating lopinavir resistance in HIV isolates harbouring CS mutation 431V and similar in subtype B and subtype G isolates. Of note, one HIV isolate harboured 431V and 431I CS mutations, which might indicate either the reversion of the substrate to wild-type specificities or the ongoing evolution of PI resistance. The N-terminal Gag CS mutation 128I was observed in this study twice in PI-resistant HIV isolates and was initially found in PI-resistant HIV isolates without PR mutations,¹¹ which were selected after PI exposure *in vitro*. Additionally, CS mutation 128I was selected *in vivo* in patients failing darunavir therapies.¹² Therefore, CS mutation 128I might influence the phenotypic PI resistance or at least lower the genetic barrier for the evolution of PI resistance in at least subtype B and G isolates.

CS mutation 453I was only detected in PI-resistant HIV isolates, suggesting an association with PI failure in this study, which is also supported by its rare occurrence in treatment-naïve HIV subtype B and non-B isolates (<1%).¹³ This mutation might be favoured in HIV isolates harbouring the polymorphism 453T instead of 453P wild-type, because CS mutation 453I is encoded by ATA and requires only a single nucleotide exchange from baseline polymorphism 453T (ACA). In contrast, CS mutation 453L (CTA), which is found in ~24% of PI-resistant HIV subtype B isolates, would only occur after two nucleotide changes. CS mutation 436R was previously found to accumulate in PI-resistant HIV subtype B isolates¹⁰ and was reported to confer low-level lopinavir resistance in phenotypic assays.⁵ Accordingly, in this study this mutation was observed in HIV isolates twice carrying major PR mutations and once from a patient failing lopinavir therapy. Additionally, CS mutation 436R was detected in one HIV isolate from a patient failing indinavir

Table 1. PR and Gag CS mutations in Russian sequences

Patient	Current therapy	Other previously used drugs	PR mutations	Gag CS mutations
PI failure with PR mutations				
14	ddI, 3TC, <u>LPV/r</u>	NVP, EFV, SQV	3I, 10I, 12K, 13V, 15V, 20I, 35N, 36I, 37D, 41K, 43T, <u>46L</u> , <u>48V</u> , <u>54V</u> , 61HN, 63C, 67W, 69K, 74S, <u>82S</u> , 89I, <u>90M</u>	132F, 362I, 376M, 380R, 381G, 430R, 436R
19	3TC, ddI, <u>LPV/r</u>	EFV	3I, 10V, 11IV, 12S, 13V, 15IV, 19P, 20I, 35DN, 36I, 37N, 41K, 45V, <u>46I</u> , <u>50V</u> , <u>53L</u> , <u>54V</u> , 61HQ, 64V, 67CY, 68D, 69K, 71V, 82M, 89I	362I, 374T, 376V, 381G, 451S
7	3TC, d4T, <u>LPV/r</u>	EFV	3I, 10V, 13A, 20IM, 22AV, 35D, 36I, 37N, 41KR, <u>46I</u> , <u>54V</u> , 58E, 62IV, 63P, 69K, <u>76V</u> , <u>82M</u> , 89I, 91S, 99L	128A, 376V, 382K, 431V
8	d4T, ddI, <u>LPV/r</u>	EFV, 3TC, AZT, IDV, NFV	3I, 4A, 10I, 13V, 15V, 19P, 20I, 35S, 36I, 37E, 41K, 43T, <u>46I</u> , <u>54A</u> , 56A, 57K, 63P, 69K, 71V, 74S, 77I, <u>82M</u> , 89I, <u>90M</u> , 92K	373T, 381G, 431V, 449I, 452K, 453I
17	NVP, ddI, <u>LPV/r</u>	SQV, EFV	3I, 10V, 13A, 20V, 22V, 33I, 35D, 36I, 37E, 41K, <u>46I</u> , <u>54V</u> , 57K, 62V, 63LP, 69K, 70R, 71T, 74S, <u>82A</u> , 89V, <u>90M</u>	128I, 375T, 431V, 436R, 451S, 452G
11	3TC, <u>IDV</u>	EFV	3I, 13V, 19P, 20I, <u>30N</u> , 33I, 35N, 36I, 37D, 41K, 43R, 45Q, <u>54V</u> , 69K, 74S, 82I, <u>88D</u> , 89V	128A, 376V, 430R, 453I
4	d4T, 3TC, <u>NFV</u>	—	3I, 13V, 20I, <u>30N</u> , 35N, 36I, 37N, 41K, 45R, 58E, 69K, 70R, 82I, <u>88D</u> , 89M	128I, 453I
2	d4T, 3TC, <u>NFV</u>	—	3I, 13V, 20V, 35D, 36I, 37D, 40EG, 41K, 69K, 74S, 82I, 89I, <u>90M</u>	362I, 381G
5	ddI, EFV, <u>NFV</u>	—	3I, 13V, 15V, 19IL, 20AITV, 35DG, 36I, 37D, 41K, 62IV, 63LP, 69K, 70KR, 71T, 74ST, 82I, 89I, <u>90M</u>	436R
6	ddI, NVP, <u>NFV</u>	—	3I, 13V, 15V, 20T, 35D, 36I, 37DE, 41K, 57K, 62V, 69K, 71AT, 74S, 82I, 89V, <u>90M</u>	132F, 376M, 430R
21	<u>NFV</u> , <u>IDV</u> , 3TC	d4T, ddI, NVP	3I, 13V, 20I, 35D, 36I, 37N, 41K, 69K, 82I, 85IV, 89M, <u>90M</u>	376V, 429N, 431VI
PI failure without PR mutations				
10	d4T, ddI, <u>NFV</u>	3TC, NVP	3I, 13V, 14KR, 20I, 35D, 36I, 37D, 41K, 69K, 70R, 82I, 89M	128T, 373T, 381G, 428Q, 430G
13	<u>IDV</u> , 3TC, AZT	ddI, NFV	3I, 13V, 20I, 35D, 36I, 37N, 41K, 43R, 69K, 82I, 89M	128T
18	<u>IDV</u> , 3TC	—	3I, 4S, 13V, 20I, 35D, 36I, 37N, 41K, 43R, 67Y, 69K, 82I, 89M	128A
20	EFV, <u>IDV</u>	—	3I, 10I, 13V, 20I, 35D, 36I, 37N, 41K, 69K, 82I, 89M, 99L	436R
RTI failure				
12	AZT, 3TC	—	3I, 10I, 13V, 20I, 35D, 36I, 37I, 41K, 66IV, 69K, 82I, 89I, 99L	373T, 376A, 380R
9	EFV, d4T, ddI	3TC	3I, 13V, 15V, 20V, 32AV, 35D, 36I, 37N, 41K, 57K, 69K, 74S, 82I, 89M	130R, 132F, 376V
16	EFV	AZT, 3TC	3I, 4S, 13V, 14KR, 20I, 36I, 37D, 41K, 43R, 64M, 67Y, 69K, 82I, 87GR, 89M	128A, 373—, 376V
15	NVP, AZT	NVP, AZT, 3TC	3I, 12AE, 13V, 20I, 35D, 36I, 37D, 41K, 43KR, 57KR, 61H, 63LV, 64M, 67CY, 69K, 82I, 89M	128A, 376V, 428D
No treatment				
3	no	—	3I, 10I, 13V, 15V, 20I, 35D, 36I, 37N, 41K, 58E, 69K, 82I, 89M, 99L	128A, 135M
22	no	AZT	3I, 12S, 13V, 14R, 15V, 20I, 35D, 36I, 37N, 41K, 69K, 82I, 89M	

3TC, lamivudine; AZT, zidovudine; d4T, stavudine; ddI, didanosine; EFV, efavirenz; IDV, indinavir; LPV/r, lopinavir/ritonavir; NVP, nevirapine; NFV, nelfinavir; SQV, saquinavir. Patients were classified according to their current antiretroviral therapy (PIs are underlined) and the detection of major PR mutations, which are also underlined.

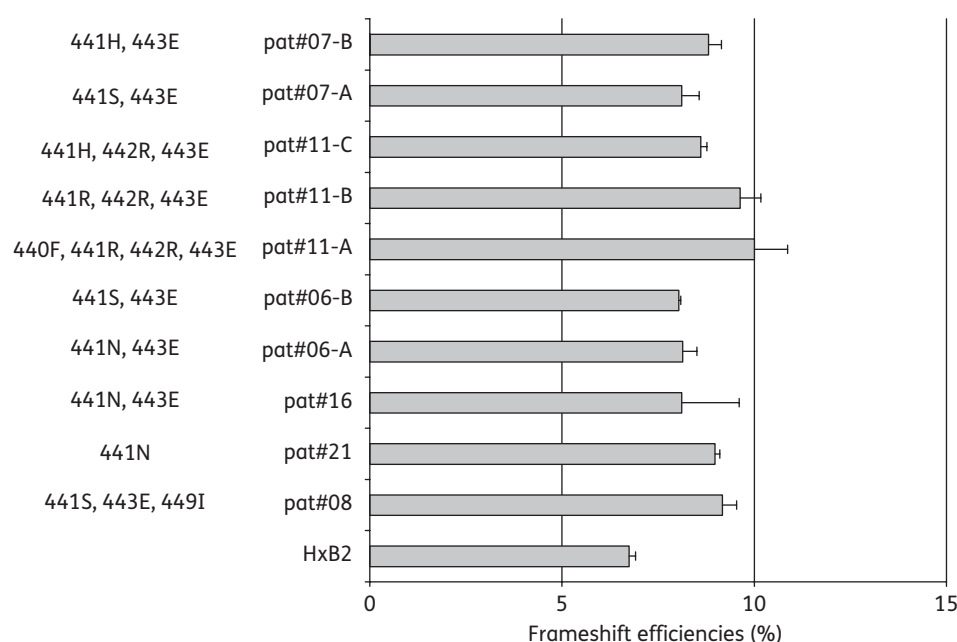


Figure 1. Frameshift efficiencies in dual luciferase assays of p1 sequences harbouring different Gag mutations (amino acids 441–449).

therapy, which did not harbour major PR mutations but another Gag mutation (123E) associated with PI resistance. The role of HIV non-CS mutations (76K, 79F and 81A) in one of these subtype G isolates from patients failing nelfinavir therapy, which were most recently described in a PI-resistant subtype B isolate,³ has to be determined in further studies.

Non-CS mutations 123E and 200I were selected *in vitro* after exposure of HIV isolates to PIs, and these HIV isolates displayed phenotypic PI resistance in the absence of PR mutations.^{2,4,11} To our knowledge, both mutations have been observed in clinical HIV isolates carrying major PR mutations for the first time in this study. The occurrence of these mutations *in vitro* and *in vivo* underlines the need for further studies clarifying the mechanism of resistance. Even though unique mutations were observed in the frameshift-regulating site, especially in PI-resistant HIV isolates, the frameshift efficiency was not affected. Therefore, the theoretical mechanism of resistance conferred by an increased amount of viral enzymes due to increased frameshift efficiency does not hold true for these HIV isolates.

In summary, Gag CS mutations were correlated with the presence of PI resistance PR mutations in these subtype G isolates, supporting the fact that in PI resistance PR and CS mutations seem to be two sides of the same coin. The impact of HIV Gag mutations in the absence of PR mutations is still a matter of debate. Since single CS mutations only conferred low-level PI resistance, the presence of one CS mutation would hardly explain the failure of boosted PI therapy regimens. However, here the failure of at least one PI therapy might be related to Gag mutations in the absence of major PR mutations.

Funding

This work was supported by the Koeln Fortune Program (number 32/2005).

Transparency declarations

None to declare.

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